

SOUTH HAVEN, MI, USA

WMO: 722343

Lat: 42.351N

Lon: 86.256W

Elev: 203

StdP: 98.91

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-12.1	-18.9	0.7	-13.7	-16.4	0.9	-10.9	10.6	-1.0	9.4	-1.7	3.0	200	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.1	22.6	29.6	21.9	27.9	21.2	24.3	28.5	23.5	27.4	22.7	26.5	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	18.1	26.6	22.3	17.4	26.0	21.4	16.4	25.1	74.5	28.6	71.0	27.5	67.8	26.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.4	-19.4	32.9	3.7	1.6	-22.1	34.0	-24.2	35.0	-26.3	35.9	-28.9	37.0		
			-19.8	25.7	3.5	1.1	-22.4	26.5	-24.4	27.2	-26.4	27.8	-29.0	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-2.8	-2.4	2.4	8.4	14.3	19.4	21.5	20.8	17.7	11.4	5.3	0.3
	DBStd	9.89	5.50	5.41	5.98	5.17	5.12	3.80	3.33	2.94	4.21	4.82	4.81	4.58
	HDD10.0	1601	398	348	248	91	16	0	0	0	1	39	157	303
	HDD18.3	3466	656	581	494	299	147	31	9	11	63	222	392	559
	CDD10.0	1508	1	1	13	43	149	283	356	333	231	82	14	2
	CDD18.3	332	0	0	1	2	22	63	108	86	42	7	0	0
	CDH23.3	2612	0	0	8	22	219	507	896	606	303	50	0	0
	CDH26.7	654	0	0	1	1	56	132	251	128	79	7	0	0
Wind	WSAvg	3.0	3.8	3.6	3.4	3.6	2.8	2.4	2.2	2.0	2.3	3.2	3.6	3.7
Precipitation	PrecAvg	742	47	34	49	73	71	70	72	77	85	65	73	57
	PrecMax	1192	90	143	109	154	200	184	269	274	321	159	205	142
	PrecMin	526	2	0	10	16	18	8	14	7	0	10	28	16
	PrecStd	129	25	26	24	31	42	41	45	50	60	35	40	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	15.4	24.0	26.1	30.7	32.3	33.0	32.0	31.3	27.5	21.1	15.9
		MCWB	13.3	12.0	15.6	15.8	21.4	22.5	24.3	22.7	22.0	20.7	14.1	13.5
	2%	DB	9.9	11.0	19.1	22.9	28.1	30.0	31.2	29.8	29.0	24.8	17.4	11.9
		MCWB	8.3	8.4	13.7	14.1	19.4	21.4	23.3	22.3	21.8	18.6	12.4	10.4
	5%	DB	6.6	7.4	15.2	20.9	25.9	27.9	29.2	27.9	27.0	22.1	15.2	8.9
		MCWB	4.7	5.1	11.0	12.7	18.1	20.9	22.3	21.8	20.5	17.5	11.1	6.9
	10%	DB	4.0	4.7	11.7	17.5	23.0	26.3	27.7	26.6	24.8	19.4	12.7	6.3
		MCWB	2.6	2.6	8.1	11.3	17.0	20.0	21.7	21.1	19.7	16.0	9.5	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.9	16.9	17.8	22.3	24.4	25.8	25.2	24.6	21.9	16.0	14.2
		MCDB	14.8	13.8	22.9	23.5	28.7	29.7	30.8	29.2	28.5	26.6	18.3	15.2
	2%	WB	8.3	8.4	14.6	15.6	20.8	23.0	24.5	23.9	23.0	20.0	13.7	10.1
		MCDB	9.1	10.7	18.1	20.4	26.1	27.6	29.1	27.4	26.8	23.2	17.0	11.6
	5%	WB	4.9	5.3	11.7	14.0	19.5	22.1	23.7	23.0	21.8	18.4	11.6	7.1
		MCDB	6.4	7.4	14.9	19.3	24.4	26.4	27.8	26.4	25.2	20.9	14.5	8.3
	10%	WB	2.9	2.8	8.0	11.9	18.0	21.0	22.8	22.2	20.6	16.0	9.8	4.6
		MCDB	4.1	4.5	11.0	16.9	22.0	24.8	26.5	25.5	23.5	19.1	12.8	6.3
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.8	8.5	10.2	10.8	10.1	9.9	9.8	10.1	9.0	7.2	5.5
		MCDBR	9.0	10.4	13.2	14.7	13.4	11.5	11.2	11.1	11.6	11.9	10.5	9.3
	5% WB	MCWBR	7.7	8.6	9.0	8.2	6.6	5.5	5.4	5.6	5.5	7.1	7.2	7.9
		MCDWR	8.4	10.1	11.9	13.0	12.1	9.5	9.7	9.0	9.4	10.5	9.5	8.8
Clear-Sky Solar Irradiance	taub		0.302	0.331	0.345	0.384	0.413	0.428	0.431	0.428	0.400	0.357	0.321	0.305
		taud	2.416	2.322	2.366	2.286	2.267	2.276	2.269	2.279	2.322	2.434	2.483	2.449
	Ebn at Noon		841	865	898	884	863	848	841	832	828	828	813	804
		Edh at Noon	78	100	108	127	133	132	132	126	113	89	73	70
All-Sky Solar Radiation	RadAvg		1.26	2.06	3.54	4.75	5.48	6.23	6.34	5.42	4.34	2.73	1.61	1.05
	RadStd		0.14	0.29	0.30	0.41	0.50	0.41	0.28	0.30	0.37	0.31	0.25	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.94	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page